

STRELKA, Russian Federation

WMO: 292740

Lat: 58.091N

Lon: 93.162E

Elev: 90

StdP: 100.25

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.2	-36.8	-42.9	0.1	-40.3	-39.0	0.1	-35.9	7.2	-14.3	6.5	-15.9	0.7	50	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	29.0	19.5	27.1	18.6	25.4	17.8	20.8	26.4	19.8	25.2	18.9	23.7	2.3	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	13.8	23.3	17.9	13.0	22.4	16.9	12.2	21.3	60.6	26.5	57.4	25.4	54.2	23.7	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	6.1	5.4	-42.5	32.2	3.2	1.3	-44.8	33.2	-46.7	33.9	-48.5	34.7	-50.9	35.7		
(5)			-41.8	22.5	3.7	1.4	-44.5	23.5	-46.7	24.4	-48.7	25.2	-51.4	26.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-21.4	-16.6	-7.7	1.7	8.6	16.2	18.9	16.2	8.4	0.4	-11.3	-15.2		
	DBStd	14.96	9.78	8.16	7.43	4.30	4.92	4.39	3.13	3.46	4.36	5.02	8.57	9.19		
	HDD10.0	4412	974	744	548	251	87	6	0	2	82	297	638	782		
	HDD18.3	6808	1233	978	806	500	304	89	31	83	299	556	888	1041		
	CDD10.0	741	0	0	0	1	44	192	276	194	33	0	0	0		
	CDD18.3	94	0	0	0	0	2	26	49	18	0	0	0	0		
	CDH23.3	874	0	0	0	1	34	295	401	142	3	0	0	0		
	CDH26.7	198	0	0	0	0	7	72	95	24	0	0	0	0		
	WSAvg	2.4	2.3	2.3	2.5	2.8	2.6	2.0	1.8	1.9	2.3	2.8	3.0	2.8		
	PrecAvg	476	25	21	19	27	44	55	52	61	51	44	41	39		
(15) Precipitation	PrecMax	714	52	31	40	38	91	110	164	118	107	81	60	60		
	PrecMin	367	3	8	0	6	5	2	7	26	10	14	17	22		
	PrecStd	94	10	7	10	8	25	28	39	26	23	18	11	9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.8	0.8	8.4	17.4	27.6	31.1	31.3	29.2	23.2	13.5	6.5	0.9		
		MCWB	-2.1	-1.2	3.3	9.0	15.9	20.7	20.8	20.8	15.5	8.4	4.8	-0.6		
	2%	DB	-3.4	-1.1	5.5	13.8	23.1	28.7	29.3	26.8	19.7	10.9	3.6	-0.6		
		MCWB	-4.5	-2.7	1.6	7.2	13.7	18.9	19.9	19.0	13.9	6.7	1.9	-1.7		
	5%	DB	-5.2	-2.8	3.7	10.7	20.0	26.8	27.3	24.8	17.2	8.7	1.9	-2.3		
		MCWB	-6.1	-4.0	0.5	5.3	11.9	17.9	19.3	18.1	12.7	5.9	0.5	-3.3		
	10%	DB	-7.5	-4.9	2.0	8.3	17.2	24.7	25.6	22.6	15.1	6.7	0.0	-4.1		
		MCWB	-8.2	-6.0	-0.4	3.9	10.7	17.0	18.2	17.3	11.8	4.7	-1.0	-5.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.0	-1.1	3.8	9.8	16.9	22.0	22.6	21.3	16.3	9.1	5.1	-0.3		
		MCDB	-0.9	0.2	7.1	16.4	25.4	28.9	28.7	27.3	21.4	12.3	6.6	0.4		
	2%	WB	-4.3	-2.5	2.0	7.5	14.5	20.2	21.1	20.1	14.6	7.3	2.1	-1.6		
		MCDB	-3.5	-1.3	4.5	12.6	21.2	26.4	26.7	24.9	18.4	10.0	3.1	-0.7		
	5%	WB	-6.1	-4.0	0.8	5.8	12.9	19.1	20.1	19.0	13.4	6.0	0.6	-3.2		
		MCDB	-5.3	-2.7	3.2	10.0	18.8	25.2	25.6	23.1	16.5	8.1	1.8	-2.3		
	10%	WB	-8.1	-6.0	-0.3	4.2	11.4	17.8	19.2	17.8	11.9	4.8	-1.0	-4.9		
		MCDB	-7.4	-5.0	2.2	7.9	16.4	23.0	24.1	21.6	14.9	6.6	0.2	-4.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.3	9.7	11.1	9.1	10.3	11.0	10.5	9.2	7.3	5.1	5.6	6.2		
		MCDBR	7.9	9.6	10.4	13.3	15.7	15.1	13.7	12.6	11.4	7.8	6.0	6.9		
		MCWBR	7.4	8.5	7.7	8.2	7.9	6.5	5.8	6.1	6.3	5.2	5.3	6.5		
		MCDBR	8.0	9.0	9.4	12.3	14.3	13.5	11.7	10.8	9.9	7.1	5.8	6.7		
	5% WB	MCWBR	7.6	8.0	7.3	8.0	7.8	6.5	5.7	5.9	6.0	5.2	5.3	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.289	0.320	0.357	0.389	0.404	0.419	0.448	0.414	0.368	0.344	0.304	0.278		
	taud		1.957	1.963	1.945	2.052	2.225	2.277	2.200	2.274	2.359	2.233	2.006	1.921		
	Ebn at Noon		493	666	756	808	827	819	781	783	765	654	468	370		
	Edh at Noon		61	97	131	140	127	122	130	112	88	75	57	46		
(44) All-Sky Solar Radiation	RadAvg		0.61	1.54	2.86	4.32	4.99	5.89	5.41	4.18	2.47	1.19	0.58	0.34		
	RadStd		0.09	0.14	0.23	0.31	0.51	0.59	0.38	0.34	0.36	0.12	0.11	0.07		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page